Yixin Chen 陈亦新

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Research code: GitHub / YixinChen-AI
Publications: Google Scholar



Research profile

I work on universal medical image segmentation and cross-modality adaptation. I also study multi-organ metabolism using whole-body PET/CT.

14 first/co-first-author papers, 7 other collaborative papers; 12 patent records (5 granted); 1 sole-authored book.

Education

Peking University, PhD in Medical Imaging Technology, expected 2027 — September 2023–present
Institute of Medical Technology, Health Science Center; Supervisor: Prof. Zhaoheng Xie

King's College London, MSc in Artificial Intelligence, with Distinction — 2019–2020

Xiamen University, BSc in Automation; second bachelor's degree in Mathematical Finance — 2015–2019

Research funding

NSFC Young Student Basic Research Project (Doctoral Student) — January 2026–December 2027
Principal investigator, grant 625B2004, RMB 300,000. Project: Multi-Organ Metabolic Information Extraction and Analysis Using Total-Body PET/CT. PKU news coverage

Selected research

Nature Communications, 2025; Co-first author [Paper] [Code]
MPUM: Modality-projection universal model for comprehensive full-body medical imaging segmentation
Modality projection generates modality-specific kernels to segment PET, CT and MRI.

IEEE TMI, 2024; First author [Paper] [Code]
PCNet: Prior Category Network for CT Universal Segmentation Model
Anatomical category prompts and a hierarchical loss guide universal CT segmentation.

MICCAI, 2024; Poster; First author [Paper] [Code]
LUCIDA: Low-Dose Universal-Tissue CT Image Domain Adaptation for Medical Segmentation
It uses segmentation reconstruction and Fourier adaptation to segment low-dose CT tissues without target labels.

EJNMMI, 2026; Accepted; First author
Whole-Body [^{18}F]FDG PET/CT Characterizes Multi-Organ Metabolic Alterations in MASLD
In **294** adults at two centres, **16** imaging features differed between metabolic dysfunction-associated steatotic liver disease (MASLD) and controls, with replication at the second centre.

Book

Deep Learning Theory and Practice Using PyTorch
深度学习理论与实战——PyTorch案例详解
Sole author. Tsinghua University Press, 2021. In Chinese. ISBN 978-7-302-56850-6.

Publications

* Equal contribution.

2026

1. **Yixin Chen**, Chuanzhi Zhou, Lijuan Guo, et al. Whole-Body [^{18}F]FDG PET/CT Characterizes Multi-Organ Metabolic Alterations in MASLD. **EJNMMI***, 2026;* Accepted.
2. Haowei Xu, **Yixin Chen**, Wanyi Fu, Hongbin Han, Zhaoheng Xie. NeuronCtrl: Geometry-Aware Safe Closed-Loop Generative Control for Neuronal Microenvironment Dynamics. **ICML***, 2026, PMLR 306:140675-140728;* Spotlight.
3. Wenrui Shao, **Yixin Chen**, Ourui Fu, Tian Li, Zhi Yang, Nan Li, Hongbin Han, Xiangxi Meng, Zhaoheng Xie. Neural Network Approximation Aided Kinetic Parameter Inference of Compartment Models in Dynamic PET. **IEEE TRPMS***, 2026.*
4. **Yixin Chen**, Yan Wang, Wenrui Shao, Zhaoheng Xie. PHATE-Net: Differentiable Pseudotime Learning for Trustworthy Disease Trajectories in PET. **CVPR Findings***, 2026, pp. 2534-2543.*
5. **Yixin Chen**. Diagnosis-Conditioned Retrieval for Structured Pathology Workflows. **MICCAI REG² challenge workshop***, 2026, LNCS 17252;* Poster.
6. **Yixin Chen**, Yizhen Peng. Acquisition-Structured nnU-Net Ensembles for Multi-Sequence Cardiac MRI Segmentation and Quantification. **MICCAI CMRSeg workshop***, 2026, LNCS 17268;* Selected for oral presentation.
7. **Yixin Chen**. Failure-Mode-Driven Multimodal Mitral Valve Segmentation for the MVAA 2026 Challenge. **MICCAI MVAA challenge workshop***, 2026, LNCS 17268;* Poster.

2025

8. **Yixin Chen**[*], Lin Gao[*], Yajuan Gao, Rui Wang, Jingge Lian, Xiangxi Meng, Yanhua Duan, Leiying Chai, Hongbin Han, Zhaoping Cheng, Zhaoheng Xie. Modality-projection universal model for comprehensive full-body medical imaging segmentation. **Nature Communications***, 2025, 16:9423.* [Code]
9. **Yixin Chen**, Yan Wang, Zhaoheng Xie. VP-SFDA: Visual Prompt Source-Free Domain Adaptation for Cross-Modal Medical Image. **Health Data Science***, 2025, 5:0143.*
10. Wenrui Shao, Yarong Zhang, Fen Du, Fangxiao Cheng, **Yixin Chen**, Xiangxi Meng, Ying Liang, Zhaoheng Xie. Neural network-aided unsupervised input function estimation for dual-time-window PET Patlak analysis. **EJNMMI Physics***, 2025, 12(1):100.*
11. W. Shao, **Yixin Chen**, Z. Wang, R. Xu, X. Meng, Z. Xie. A Physiology-Supervised Neural Operator Framework for Kinetic Parameter Estimation in Dynamic PET. **IEEE NSS/MIC/RTSD***, 2025, p. 1.*

2024

12. **Yixin Chen**, Yajuan Gao, Lei Zhu, Wenrui Shao, Yanye Lu, Hongbin Han, Zhaoheng Xie. PCNet: Prior Category Network for CT Universal Segmentation Model. **IEEE TMI***, 2024, 43(9):3319-3330.* [Code]
13. **Yixin Chen**, Xiangxi Meng, Yan Wang, Shuang Zeng, Xi Liu, Zhaoheng Xie. LUCIDA: Low-Dose Universal-Tissue CT Image Domain Adaptation for Medical Segmentation. **MICCAI***, 2024, LNCS 15008:393-402;* Poster. [Code]
14. **Yixin Chen**[*], Yajun Gao[*], Lei Zhu, Jianan Li, Yan Wang, Jiakui Hu, Hongbin Han, Yanye Lu, Zhaoheng Xie. Structure-Enhanced Unsupervised Domain Adaptation for CT Whole-Brain Segmentation. **IEEE TRPMS***, 2024, 8(8):926-938.* [Code]
15. W. Shao, Y. Zhang, **Yixin Chen**, X. Meng, Y. Liang, Z. Xie. Neural Network-Asissted Dual-time-window Acquisition: Modeling Incomplete Input Functions for Patlak Analysis. **IEEE NSS/MIC/RTSD***, 2024, p. 1.*

2023

16. Yan Wang, Jian Cheng, **Yixin Chen**, Shuai Shao, Lanyun Zhu, Zhenzhou Wu, Tao Liu, Haogang Zhu. FVP: Fourier Visual Prompting for Source-Free Unsupervised Domain Adaptation of Medical Image Segmentation. **IEEE TMI***, 2023, 42(12):3738-3751.*
17. Yan Wang[*], **Yixin Chen**[*], Yingying Zhang, Haogang Zhu. Rethinking Disentanglement in Unsupervised Domain Adaptation for Medical Image Segmentation. **IEEE EMBC***, 2023, pp. 1-6.*
18. Xixiang Lin[*], Feifei Yang[*], **Yixin Chen**[*], et al. Echocardiography-based AI for detection and quantification of atrial septal defect. **Frontiers in Cardiovascular Medicine***, 2023, 10:985657.*

2022

19. Yan Wang[*], **Yixin Chen**[*], Wenjun Wang, Haogang Zhu. MSGAN: Multi-Stage Generative Adversarial Networks for Cross-Modality Domain Adaptation. **IEEE EMBC***, 2022, pp. 520-524.*
20. Jianan Li[*], **Yixin Chen**[*], Minyu Zhang, et al. A deep learning model system for diagnosis and management of adnexal masses. **Cancers***, 2022, 14(21):5291.*
21. Xixiang Lin, Feifei Yang, **Yixin Chen**, et al. Echocardiography-based AI detection of regional wall motion abnormalities and quantification of cardiac function in myocardial infarction. **Frontiers in Cardiovascular Medicine***, 2022, 9:903660.*

Teaching experience

Teaching assistant, Peking University: Principles of Medical Imaging Equipment

Computer vision course instructor: Independently recorded introductory and intermediate courses in computer vision.

Professional experience

AirDoc, Algorithm Engineer — July–September 2022

Developed ICL vault prediction models; worked on ultra-widefield fundus image analysis and medical device registration.

BioMind, Medical Imaging Algorithm Engineer — November 2020–June 2022

Developed cardiac ultrasound assessment and quantification models, supported clinical integration, and contributed to an ovarian-mass assessment system.

MICCAI challenge awards

PKU news coverage

autoPET V: second place in Category 1	2026
CMR-MULTI: second place	2026
REG²: fourth place	2026
CHIMERA-agent: Innovative Agent Solution Award	2026

Patents

12 public patent records: 5 granted Chinese invention patents, 6 published Chinese invention applications, and 1 published PCT international application.

Granted Chinese invention patents

1. Brain-region segmentation method and system for CT brain images
   Yixin Chen (陈亦新), 谢肇恒; CN117649519B (2026-09-11).
2. Method and apparatus, device, and medium for generating two-dimensional cardiac ultrasound view images
   蒲海涛, 陈晓天, 张培芳, **Yixin Chen (陈亦新)**, 叶菁, 高雨霏, 吴振洲; CN115511703B (2023-05-09).
3. Model-training method and apparatus, electronic device, and storage medium
   Yixin Chen (陈亦新), 王焱, 陈晓天, 任艺柯, 张培芳, 吴振洲; CN115099293B (2023-04-18).
4. Method and apparatus for removing artifacts from cardiac ultrasound images
   蒲海涛, 张培芳, 陈晓天, **Yixin Chen (陈亦新)**, 叶菁, 吴振洲; CN115170424B (2023-04-07).
5. Method and apparatus, electronic device, and storage medium for determining the length of an atrial septal defect
   Yixin Chen (陈亦新), 蒲海涛, 林锡祥, 陈晓天, 张培芳, 吴振洲; CN114565622B (2023-04-07).

Published Chinese invention applications

6. Method and system for analyzing metabolism in an organism
   Yixin Chen (陈亦新), 谢肇恒, 徐若妍; CN120458609A (2025-08-12).
7. Neural-network-based method for biomedical parametric imaging and pharmacokinetic analysis
   邵闻睿, 谢肇恒, 孟祥溪, **Yixin Chen (陈亦新)**; CN119811674A (2025-04-11).
8. Method for analyzing differences in fundus images and related products
   Yixin Chen (陈亦新), 赵昕; CN119379577A (2025-01-28).
9. Method for processing ultra-widefield fundus images and related products
   Yixin Chen (陈亦新), 赵昕; CN119379707A (2025-01-28).
10. Method for training an image-classification model and related products
    Yixin Chen (陈亦新), 赵昕; CN118736653A (2024-10-01).
11. Data-processing system and method
    焦文沛, 谢肇恒, **Yixin Chen (陈亦新)**; CN118039082A (2024-05-14).

Published PCT international application

12. Image-processing method and apparatus, electronic device, and storage medium
    齐秀梅, **Yixin Chen (陈亦新)**, 赵昕; WO2024216763A1 (2024-10-24).

Academic service

Nature Communications, Early Career Co-Reviewer